

Name: Riya Pawaskar	Roll no.: 27
Class: SYIT	Date of submission : 12/08/26

Practical 5

- a) Write a python code to demonstrate importing pandas libraries and create data frame object.

Program:

```
import pandas as pd
data = {
    "name": ["Sapna", "Nikita", "Jasad"],
    "age": [18, 19, 20],
    "qualified": [True, True, False]
}
df = pd.DataFrame(data)
newdf = df.filter(items=["name", "age"])
print(newdf)
print("Riya_27")
```

Output:

```
   name  age
0  Sapna   18
1  Nikita  19
2   Jasad  20
Riya_27
```

b) Write a python code to show statistical information on given data set.

Program:

```
import statistics
print("Mean:")
print(statistics.mean([-11, 5.5, -3.4, 7.1, -9, 22]))
print("\nMedian:")
print(statistics.median([1, 3, 5, 7, 9, 11]))
print("\nMode:")
print(statistics.mode(['red', 'green', 'blue', 'red']))
print("\nStandard Deviation:")
print(statistics.stdev([1, 30, 50, 100]))
print("\nVariance:")
print(statistics.variance([1, 3, 5, 7, 9, 11]))
print("Riya_27")
```

Output:

```
Mean:
1.8666666666666667

Median:
6.0

Mode:
red

Standard Deviation:
41.67633221226007

Variance:
14
Riya_27
```

c) Write a python code to create pandas series from dictionaries.

Program:

```
import pandas as pd
dictionary = {'DS':10, 'PDS':20, 'AM': 30}
series = pd.Series(dictionary)
print(series)
print("Riyaa_27")
```

Output:

```
DS      10
PDS     20
AM      30
dtype: int64
Riyaa_27
```

d) Write a python code to demonstrate filter pandas series with Boolean arrays.

Program:

```
import pandas as pd
dict = {'name':["Preeti", "jay", "Madhu", "sheeku"],
        'degree': ["MBA", "BCA", "B.Tech", "MBBA"],
        'score':[80, 70, 90, 78]}
df = pd.DataFrame(dict, index = [0, 1, 2, 3])
print(df[[False, True, True, False]])
print("Riyaa_27")
```

Output:

```
   name degree score
1   jay   BCA     70
2 Madhu B.Tech     90
Riyaa_27
```